DOCKHUB · CALCULATION NOTE

DockHub sizing calculations

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-31	Amish Chadha	Independent review at the kit 1.9.1 upgrade (DKH-STM-001): sensitivities added to section 11 (anchor tension with matching factors 4.13 kN, pad edge pressure); results unchanged
0.6	2026-10-02	Amish Chadha	Approved follow-ups: sizing.py rerun against the restated R6 and R9; flattest-pack solar switching as a sensitivity; R9 limits stated; anonymous prepaid cards in the token list; anchors and site pad chosen and checked (R15 met); cost with anchors and pad (USD 1,519)
0.5	2026-10-02	Amish Chadha	Requirement table and summary: R6 and R9 against the targets restated on 2026-10-02 (DKH-DEC-001); no figure changed
0.4	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.3	2026-10-01	Amish Chadha	Constructable design (DKH-DDR-003). Mass, wind, vent path, canopy height and cost rerun; R16 now not met ($1,214)

0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (charging, swap time, throughput and solar day model, grid, thermal, fire timing, geometry, mass, wind and anchoring, cost) against SwapCell interface v0.3

DockHub sizing calculations

On paper, the four-bay DockHub meets eleven of its sixteen requirements. **One is over its value-engineering target:** R16, where the anchors and site pad chosen on 2026-10-02 (USD 305) join the parts added to make the design buildable, bringing the estimated cost of the two-bay prototype to USD 1,519 against the USD 1,200 target. Two are **at risk**: R8 (fire containment, which only a propagation test can show) and R9 (against the target restated on 2026-10-02, full-rate charging up to about 30 °C ambient, the warmest bay may pause just under 30 °C). R7 and R14 cannot be verified until hardware exists. Against the targets Amish restated on 2026-10-02 (DKH-DEC-001), R6 is met with a thin margin (9.3 % against about 9 %), and R15 is now met: four M12 wedge anchors and a 1,400 x 1,100 x 300 mm cast pad hold the cabinet at a 30 m/s gust, subject to confirming the anchor rating from the maker's data.

Two findings change the TRL 2 picture. First, a swap needs an empty bay for the returned pack, so a station with *n* bays holds *n* - 1 packs. The four-bay station has three working packs (36 swaps a day at most, released full), and the two-bay first prototype has one, which serves only 9 of the 20 duty-case swaps. Second, the cabinet with its canopy frame weighs about 217 to 247 kg, not 150 to 170 kg.

Every number here is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the geometry parameters from `cad/src/model.py` and the costs from `bom/bom.csv`. All values are first-principles estimates; nothing is measured.

1. Assumptions

Table 1. Inputs. All are assumptions for a paper design.

Input	Value	Basis
Pack	13S2P, 46.8 V nominal, up to 54.6 V, 10 Ah, 468 Wh, 110 mΩ, 2.85 kg, 393 mm overall; charge window 0 to 45 °C	SwapCell interface v0.3, SWC-CAL-001
Pack heat path	1.17 W/K to the surrounding air, 32 min time constant	SWC-CAL-001 (open air)
Charger	5 A CC to 85 %, then CV tapering linearly to 1 A (0.6 h); 90 % efficient; 303 W input at the end of CC	SWC-CAL-001; certified charger
Cell charge efficiency	95 % for energy accounting; heat in the pack between I^2R (2.75 W) and all of the 5 % loss (13.7 W)	SWC-CAL-001
Duty case	20 swaps a day, packs returned at 15 %, released at 80 % or more (fullest first)	DKH-REQ-001; release threshold decided by Amish, 2026-09-25 (DKH-DDR-002)
Demand profile	Arrivals at 07:30, 08:30, 09:30, 11:00, 11:45, 12:15, 12:45, 13:15, 14:00, 15:30, 16:30, 17:30, 18:00, 18:30, 19:00, 19:30, 20:00, 20:30, 21:15 and 22:30; a rider leaves after waiting 15 min	Lunch and dinner peaks; to be tested with riders
Cabinet loads	30 W average, 40 W peak	Controller, modem, display, lights, fans, locks
Charger power factor	0.95 with power factor correction, 0.6 without	Typical ranges; the chosen charger must be checked
Branch circuit	120 V 15 A or 230 V 10 A, continuous load 80 % of the rating	Common practice for continuous loads
Solar	400 W, 4.5 peak sun hours on a sine profile from 06:00 to 18:00, 0.8 derate, MPPT 96 %, 50 V mean charging voltage; bay 1 uses the MPPT when it offers 50 W or more, otherwise its AC charger	Clear day in a sunny city; control rule for the MPPT to bay 1 decided by Amish, 2026-09-25 (DKH-DDR-002)
Cabinet heat transfer	Film coefficients 17 W/(m² K) outside and 8 W/(m² K) inside; two 120 mm fans at 90 m³/h free air each, half lost to the filter and the vent hood	Still air with light wind; typical fans
Sun on the cabinet	500 W/m² on the front and one end wall, absorptance 0.35 (light powder coat); roof shaded by the canopy	Afternoon sun
Wind	30 m/s gust, air 1.225 kg/m³; panel force coefficient 1.5 normal to the panel with the centre of pressure a quarter of the panel depth from its centre; body drag coefficient 1.3; posts 2.0; load factor 1.5 for the anchors; friction 0.4	TRL 2 assumption for the panel; typical values otherwise
Steel	7,850 kg/m³, 1.5 mm sheet, 15 % added for hems and flanges	Decided sheet gauge (R10)

2. Charging (R3)

With the SwapCell charge profile, a pack charges from 20 to 80 % in **1.20 h** and from empty to full in **2.30 h** (1.70 h CC and 0.60 h CV). R3 (1.5 h and 2.5 h) is met. A pack returned at 15 % reaches the 80 % release level in 1.30 h and full in 2.00 h.

Each full recharge from 15 % stores 397.8 Wh, takes 418.7 Wh from the charger and 465.3 Wh from the grid, about $0.093 of electricity at an assumed $0.20/kWh. Each charger draws 303 W at the end of CC and loses about 30 W as heat.

3. Swap time (R1)

Table 2. Task analysis for a rider who knows the station.

Step	Time (s)
Tap card or phone; token checked offline	3
Empty bay door unlocks	1
Open door, place the flat pack connector down	8
Close door	2
Controller confirms the return (door switch, coded INTERLOCK, pack identity on CAN)	2
Second door unlocks; open it and lift out the charged pack	10
Close door, step away	5
Total	**31**

R1 (60 s) is met with a margin of about half. The 2 s return check depends on how fast a sleeping pack wakes on the coded INTERLOCK loop and answers the heartbeat; SwapCell interface v0.3 gives no wake time, so this is an assumption.

4. Throughput and solar share (R4, R6)

One bay is always empty. A swap starts by placing the flat pack in an empty bay, so a station with *n* bays runs with *n* - 1 packs. The ceiling is (*n* - 1) x 24 h divided by the recharge time: **36 swaps a day** for four bays with packs released full, or 55 at the 80 % release level; **12** (or 18) for the two-bay first prototype. At a peak, four bays can serve about 2.3 swaps an hour after the first three; two bays serve 0.77.

A minute-step model of the design day (the script runs three identical days and reports the third) gives Table 3.

Table 3. Design day, 20 swaps, packs returned at 15 %.

Case	Served	Longest wait	Solar used	Solar share of charging energy
Four bays, fullest pack released first	20 of 20	4 min	0.74 of 1.38 kWh	**9.3 %**
Four bays, bay 1 released first when the sun is up	19 of 20	0 (one rider leaves after 15 min)	0.86 of 1.38 kWh	11.4 %
Two bays (first prototype)	9 of 20	0 (11 riders leave)	0.69 of 1.38 kWh	19.9 %

R4 is met by the four-bay fit: all 20 riders are served and only one waits, for 4 minutes. The two-bay prototype, adopted to meet the value-engineering target (DKH-DDR-001 item 1), can show the swap, charge and safety chain but cannot run the duty case.

R6 is met with a thin margin. The panel gives 1.44 kWh of DC on the design day and the MPPT delivers 1.38 kWh, but it can feed only bay 1. Bay 1 is sometimes the empty bay and at other times holds a pack that is already full, so only 0.74 kWh reaches a pack: 9.3 % of the charging energy against the target of about 9 % with normal routing restated on 2026-10-02 (it was 10 %). Sending flat packs to bay 1 in daylight raises the share to 11.4 %, but solar charging at up to 181 W is slower than the 5 A AC charger, so fewer packs are ready at the dinner peak and one rider in 20 leaves. The TRL 2 estimate of 13 % assumed all the solar energy would be used. The peak MPPT output of 181 W is below the 5 A bay limit (about 250 W), so there is no clipping on the design day.

Sensitivity: solar switched to the flattest pack. Amish decided on 2026-10-02 that switching the solar charger to whichever bay holds the flattest pack is to be tried as a TRL 4 controller change, not a cabinet change. As a sensitivity only, the script also runs that rule (the solar output follows the flattest pack still charging, and that bay's AC charger rests while it does): all 20 riders are served with no wait and the solar share rises to **16.1 %** (1.28 kWh of the 1.38 kWh available). The rule is not adopted; R6 is assessed with normal routing, and the share is rerun if the change is adopted after the TRL 4 trial.

5. Daily energy

At the duty case the four-bay station takes **8.76 kWh a day from the grid** (8.04 kWh to the chargers and 0.72 kWh for cabinet loads) and 0.77 kWh from the panel, and stores 7.58 kWh in the packs riders take away. Losses are 0.80 kWh in the chargers, 0.03 kWh in the MPPT and 0.40 kWh in the cells. Figure 2 of DKH-PRC-001 uses these values. Grid energy per swap, including the cabinet loads, costs about $0.088 at $0.20/kWh.

6. Grid supply (R5)

With four chargers at the end of CC and 40 W of auxiliaries, the peak real power is **1,253 W**, inside the 1.5 kW target (647 W with two bays fitted). In the day model at most three chargers run at once, because one bay is always empty.

Table 4. Supply current with four chargers running.

Charger power factor	Apparent power	120 V (12 A continuous limit)	230 V (8 A continuous limit)
0.95 (with correction)	1,317 VA	11.0 A	5.7 A
0.6 (without)	2,062 VA	**17.2 A**	**9.0 A**

R5 is met **only with chargers whose power factor is 0.9 or better**; without power factor correction four chargers overload a 120 V 15 A circuit and exceed the continuous limit of a 230 V 10 A circuit. `bom/bom.csv` line 5 now specifies this.

7. Thermal (R9)

The cabinet skin has an area of 5.20 m², a U-value of 5.44 W/(m² K) and a loss conductance of 28.3 W/K; the fans move 90 m³/h (53 cfm), worth 30.1 W/K. With three packs charging, the heat inside is 91 W from the chargers, 8 to 41 W from the packs, 30 W of cabinet loads and 118 W of sun through the walls. The cabinet air rises only **4.2 to 4.8 K** above ambient, less than the 10 K assumed at TRL 2, so at 45 °C ambient it is about 49 °C, not 55 °C.

That does not rescue R9, because the pack itself refuses to charge above 45 °C. By the end of the CC phase a pack is 2.2 to 10.8 K above the cabinet air, so **uninterrupted 5 A charging needs ambient air below about 29 to 39 °C**. At the 45 °C design ambient the packs are at 51 to 61 °C and do not charge at all; a fixed 5 A charger can only pause. At the cold end, an idle cabinet runs only 1.1 K above ambient with the fans off, so a pack that has cooled in the cabinet cannot start charging below about -1 °C; a pack returned warm from a ride can. The requirement that the station never charges outside the pack's window is met by the pack and the controller, but the original operating range of -10 to 45 °C for charging is not.

R9 as restated on 2026-10-02 (DKH-DEC-001). Amish kept the pack's 45 °C charge limit and accepted daytime pauses. The restated limits are: the station stores packs and serves swaps at -10 to 45 °C ambient; it charges at the full 5 A rate up to about **30 °C ambient**, and above that it may pause charging in the daytime; it does not start charging a cold-soaked pack below about **-1 °C ambient** (calculated -1.1 °C); and it never charges a pack outside the pack's reported window. Against these limits the calculated full-rate limit of about 29 to 39 °C leaves the warmest bay, with the upper bound on pack heat, able to pause just under 30 °C, so **R9 is at risk** rather than met. The first pilot site is chosen with summer highs near 30 °C; active cooling is revisited only for a hot-climate site.

8. Fire detection and venting (R7, R8)

The electronic part of the detection chain takes about **3.1 s**: a 1 s NTC sample period, two confirming samples and a 0.1 s controller loop, leaving 6.9 s of the 10 s target for the sensor's own thermal lag. A pack's over-temperature fault arrives on CAN as soon as it changes. Opening the bay contactor takes about **0.15 s** (0.10 s decision and 0.05 s drop-out), inside the 1 s target. Smoke transport to the plenum detector and the sensor lag can only be measured, so R7 is **not verifiable at TRL 3**.

The vent path widens from each 36 cm² bay port into a 1,366 cm² plenum, through the two roof fans and out through 240 cm² of slots in the back face of the vent hood, where the fans move air at about 1.0 m/s. Whether this path, two 1.5 mm steel walls and the air gap stop a runaway from spreading for 30 min (R8) can be shown only by a propagation test, which is TRL 4 work and on hold. R8 stays **at risk**.

9. Geometry (R2, R11, R13)

The model places the pack's connector face on the cradle at 730 mm and its handle top at 1,105 mm. The door opening is 510 x 150 mm (narrowed from 172 mm to leave room for the door hinges, DKH-DDR-003) against the 393 mm overall pack, leaving 110 mm above the handle for the hand and 30 mm each side of the pack; the rider lifts the pack 19 mm to clear the receptacle. Every operable part lies between 0.73 and 1.10 m (door handles 0.94 to 0.98 m, reader 0.90 to 0.99 m), inside the 0.38 to 1.22 m range: **R11 met**.

The plinth is now flush with the body, so the footprint is 1,000 x 500 mm (door handles add 22 mm at the front); at TRL 2 it was 1,020 x 520 mm, which broke the 1.0 m width limit. The panel's rear edge underside is at 2,214 mm and its front top edge at 2,446 mm; the lowest canopy point over the sidewalk is the rear end of a rail at 2,171 mm (printed by `cad/src/model.py`). **R13 met.**

For R2, each bay builds to SwapCell interface v0.3 without changing it: a blind-mate receptacle with a 10 kΩ coding resistor in the INTERLOCK loop (item W), dock host type 1 in the heartbeat, a class D gravity catch for the pack's latch pawl (item V) and a CAN channel per bay at 250 kbit/s. At 34.1 frames per second one pack loads its channel to 1.8 %; the four SPI CAN controllers need about 17.5 kbit/s of SPI traffic in all. Packs still default to node 0, and v0.3 node numbers 0 to 7 do not remove the need for separate channels, because a pack's node number would have to change at every station. **R2 met by design review.**

For R12, a 64 B log record per swap is 1.28 kB a day, and a list of 10,000 rider tokens at 20 B each is 200 kB, both trivial for the controller's flash and SD card over 24 h offline. Each token entry holds an 8 B hash of the card's identifier, a type (named token or anonymous prepaid card, decided 2026-10-02), whether a pack deposit is held, the prepaid swaps left, an expiry date and flags; no name or contact detail is stored. **R12 met by design review.**

10. Mass

Table 5. Mass of the four-bay station with three packs.

Part	Mass (kg)
Sheet steel, 10.24 m² at 1.5 mm plus 15 %	138.6
Plinth, 4.0 m of 100 x 50 x 5 mm channel	29.8
Canopy posts and rails	21.0
Roof beams and cleats, post base and cap plates, hinges, end clamps, catch brackets	9.6
Solar panel	21.0
Chargers (4)	4.8
Cradles and receptacles (4)	2.0
Controller, MPPT, grid unit, access panel, fire unit, fans, wiring	12.0
Packs (3)	8.6
Total	**247**

The two-bay prototype without packs weighs about **217 kg**, which is used for the wind check. The site pad chosen on 2026-10-02 adds about 1,109 kg of concrete under it (section 11); the trial vermiculite trays add well under 1 kg a bay. The TRL 2 estimate of 150 to 170 kg counted about 8 m² of sheet; the model's bay liners, deck, plenum wall, charger shelf and vent hood bring it to 10.24 m². The roof frame, post plates, hinges and other parts added to make the design buildable (DKH-DDR-003) add 9.6 kg.

11. Wind and anchoring (R15)

At a 30 m/s gust the dynamic pressure is 551 Pa. The 1.95 m² panel takes a normal force of 1,615 N, which at 10° of tilt is 280 N sideways and **1,590 N of uplift**, acting 429 mm in front of the cabinet centre; the posts add 158 N and the body 1,075 N of drag. About the rear bottom edge this gives an overturning moment of **2.91 kN·m**, against 0.53 kN·m from the prototype's own weight: a factor of 0.18. Unanchored, the cabinet would tip, and friction alone (215 N) would not stop it sliding either. The TRL 2 note treated the canopy force as horizontal and missed the uplift; the corrected moment is lower than its 3.6 kN·m, but the conclusion stands.

With four M12 anchors 30 mm in from the front and rear edges (and 25 mm in from the ends, where a socket reaches them through capped holes in the floor), each front anchor carries **2.53 kN** of tension, **3.79 kN** with the 1.5 load factor, and each anchor about 0.57 kN of design shear. Each front canopy post pulls up on the roof frame with about 1,251 N, so each post bolts through the roof sheet into a 40 x 40 x 4 mm angle roof beam, which is bolted at each end to the front and back panels (DKH-DDR-003); the posts themselves see only 9.4 MPa in bending (50 x 50 x 3 mm hollow section, 714 mm long). **Anchors and pad (chosen 2026-10-02).** The first pilot is on private ground, so

the cabinet gets its own cast pad rather than relying on the host's ground: reinforced concrete 1,400 x 1,100 x 300 mm (about **1,109 kg**), top flush with the ground, the cabinet centred on it, and four M12 x 160 mm stainless wedge anchors at 100 mm embedment (BOM lines 17 and 18). The anchors sit 225 mm from the pad's ends and 330 mm from its front and back edges. Taking the cabinet and pad together about the pad's rear bottom edge, the gust gives an overturning moment of **3.84 kN·m** against **7.15 kN·m** of restoring moment; with a factor of 0.9 on the weight and 1.5 on the wind the ratio is **1.12**, above the 1.0 needed. Friction under the pad (coefficient 0.5) resists 5,708 N against 2,269 N of factored sliding force, and the bearing pressure is only 8.4 kPa. The anchors still carry 3.79 kN of design tension each, which must be at or below the design resistance in the chosen anchor's approval data. A socket reaches each anchor nut through the floor and a 32 mm hole in the plinth's top flange (added on 2026-10-02; the flange had blocked it). **R15 is met** on paper, subject to that confirmation.

Sensitivity from the independent review (2026-10-31, not adopted; `sizing.py` *prints it).* Applying the pad check's own factors (1.5 on wind, 0.9 on self-weight) to the anchors gives **4.13 kN** design tension each, 9 % above the 3.79 kN above; asking the maker for a design resistance of at least 4.2 kN is proposed, awaiting Amish (DKH-DEC-001 O5). The pad's resultant sits 290 mm from its rear edge, outside the middle third (from 367 mm), so the edge pressure is about **19 kPa** unfactored and 61 kPa factored, not the 8.4 kPa average; the soil bearing capacity is to assess at the site. Only wind from the front is checked; other directions, the canopy fixings and the panel clamps are open items.

12. Cost (R16)

Table 6. Cost summary from `bom/bom.csv` *(two of four bays fitted).*

Group	Lines	Cost (USD)
Cabinet, doors, service door, plinth	1, 2, 14, 15	379
Bays (2): compartments, chargers	3, 5	180
Controller, access, grid, wiring	6, 7, 8, 16	250
Fire detection and venting	9, 10	155
Solar canopy and MPPT	11, 12, 13	250
Site: anchors and pad	17, 18	305
Total	1 to 18	**1,519**

The total is $319 over the $1,200 value-engineering target (`budget_usd` , a hypothetical control target, not a limit), so R16 is **over the target** on indicative prices. At v0.2 it was $1,199; the canopy end clamps and post plates (line 12, $10) and the service door hinge (line 15, $5) added to make the design buildable (DKH-DDR-003) took it to $1,214, and the anchors ($40) and cast pad ($265) chosen on 2026-10-02 add $305. The vermiculite floor tray (line 19) is a trial option at about $11 a bay and is not in the total. Cost drivers and savings worth trying are in the Value engineering section of the design decisions register (DKH-DEC-001). Each further bay adds $112; all four bays fitted cost $1,743. SwapCell packs are not in the station cost: the prototype needs one resident pack and the four-bay station three, at about $414 each in the SwapCell BOM, plus one per rider in circulation.

13. Results against requirements

Table 7. Requirement status. Not met, at risk and not verifiable items first.

ID	Value (DKH-CAL-001)	Target	Status
R16	$1,519 with two of four bays fitted, anchors and pad included ($1,743 with four)	$1,200, packs excluded	**Over the value-engineering target by $319**
R9	Full-rate charging up to about 29 to 39 °C ambient; cold-soaked packs start only above about -1 °C; charge window enforced	Full-rate charging up to about 30 °C ambient; cold-soak limit near -1 °C stated; never charge outside the pack window (restated 2026-10-02, DKH-DEC-001)	At risk (the warmest bay may pause just under 30 °C)
R8	Two 1.5 mm steel walls and a gap between packs; vent path to the roof	No spread for 30 min	At risk (not verifiable at TRL 3)
R7	Electronic latency 3.1 s; contactor 0.15 s	Detect in 10 s; contactor in 1 s	Not verifiable at TRL 3
R14	All parts reached through the service door or a bay door with hand tools	15 min per part	Not verifiable at TRL 3
R1	31 s task analysis	60 s or less	Met
R2	Interface v0.3: coded INTERLOCK receptacle, host type 1, class D catch, one CAN channel per bay	Build to the SwapCell interface, no change	Met (design review)
R3	20 to 80 % in 1.2 h; full in 2.3 h	1.5 h; 2.5 h	Met
R4	Four bays: 20 of 20 served, longest wait 4 min; ceiling 36 a day. Two-bay prototype: 9 of 20	20 a day from 4 bays	Met (four bays)
R5	1.25 kW; 11.0 A at 120 V with power factor 0.95 (17.2 A at 0.6)	1.5 kW; one branch circuit	Met (power factor 0.9 or better)
R6	9.3 % of charging energy (0.74 of 1.38 kWh used); 11.4 % with solar steering, which turns one rider away; 16.1 % if the solar output is switched to the flattest pack (TRL 4 trial)	About 9 % with normal routing (restated 2026-10-02, DKH-DEC-001)	Met (thin margin)
R10	1.5 mm steel, fail-secure locks, anchored plinth, token plus returned pack	Security features	Met (design review)
R11	Operable parts 0.73 to 1.10 m	0.38 to 1.22 m	Met
R12	1.28 kB of log a day; 200 kB token list with anonymous prepaid cards	No cameras; 24 h offline	Met (design review)
R13	1.0 x 0.5 m; lowest canopy point over the sidewalk 2.17 m	1.0 x 0.6 m; 2.1 m	Met
R15	Overturning 2.9 kN·m against 0.53 kN·m self-weight; four M12 anchors at 3.8 kN design tension into a 1,109 kg pad, equilibrium factor 1.12	Upright at a 30 m/s gust	Met (anchor rating to confirm from the maker's data)

Counts: 1 over its value-engineering target (not met), 2 at risk, 2 not verifiable at TRL 3, 11 met.

14. Checks against earlier documents

The TRL 2 figures in DKH-PRC-001 v0.2 and DKH-REQ-001 v0.2 were checked against this script and corrected in v0.3:

- Maximum throughput 38 to 36 swaps a day (three working packs, not four); two bays 19 to 12 a day.
- Solar share 13 % to 9.3 % (bay 1 cannot always use the sun); no clipping on the design day.
- Daily energy at 20 swaps 10.1 kWh in and 8.0 kWh stored to 9.53 kWh in (8.76 grid, 0.77 solar) and 7.58 kWh stored, because packs leave at 80 % or more.
- Swap time 30 s to 31 s with the return check.
- Cabinet air rise 10 K to 4.2 to 4.8 K; the R9 shortfall is now set by the pack's 45 °C charge limit rather than by the cabinet air.
- Mass 150 to 170 kg to 205 kg (prototype, no packs) and 234 kg (four bays, three packs); at v0.3, with the constructable design, 217 kg and 247 kg.
- Wind: 3.6 kN·m to 2.91 kN·m with the panel force resolved into uplift and drag, and anchor loads now given.
- Footprint 1,020 x 520 mm to 1,000 x 500 mm (plinth made flush).
- SwapCell pack price $370 to $414 (SWC-CAL-001).

At v0.3 the design was made constructable (DKH-DDR-003): mass 205 to 217 kg, front anchor design tension 3.83 to 3.79 kN, lowest canopy point 2.18 to 2.17 m, vent outlet 158 to 240 cm², cost $1,199 to $1,214 (R16 at risk to over the value-engineering target). No other result changed.

At v0.6 (approved follow-ups of 2026-10-02): R6 not met to met against the restated target of about 9 %; R9 not met to at risk against the restated target; R15 at risk to met with the anchors and pad chosen; R12 token list 160 to 200 kB; cost $1,214 to $1,519 with the anchors and pad; all four bays $1,438 to $1,743. No other figure changed.

> **Safety:** These are paper estimates for a cabinet that stores and charges lithium-ion packs of about 468 Wh beside a public walkway from mains power. They do not replace an electrician's design of the supply, a structural check of the anchors and pad, or a propagation test. No cabinet may be built, powered or anchored in a public place from this note; building and testing are TRL 4 work and on hold by Amish's instruction.